



Digital Egypt Pioneers Initiative (DEPI)

Pregnancy Care App

AI-Powered Maternal Health & Pregnancy Risk Intelligence Platform

Software Requirement Specification (SRS)

Data Science & Artificial Intelligence Track

Graduation Project – DEPI 2026

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1. Introduction

The **Maternal Care Platform** is an AI-powered web application developed to support pregnant women by providing intelligent pregnancy monitoring and early risk assessment. The platform combines **Machine Learning**, **Optical Character Recognition (OCR)**, and an **AI Health Assistant** to help users make informed decisions throughout their pregnancy journey.

Users can manually enter their health information or upload medical laboratory reports for automatic data extraction and pregnancy risk prediction. The platform also provides pregnancy tracking, personalized health guidance, educational resources, and a history of previous predictions.

The system is developed using **Python** and **Streamlit**, with a **Random Forest** model for pregnancy risk prediction, **EasyOCR** for medical report analysis, and **SQLite** for secure data storage. Its goal is to improve pregnancy awareness by providing fast, accessible, and easy-to-use healthcare support while emphasizing that all predictions and recommendations should be confirmed by qualified healthcare professionals.

1.2 Product Scope

The Maternal Care Platform provides an integrated solution for pregnancy monitoring and maternal health support through Artificial Intelligence. The platform enables users to:

- Register and securely log into the system.
- Predict pregnancy risk using health information.
- Track pregnancy progress with weekly milestones.
- Upload medical reports for automatic OCR analysis.
- Receive AI-assisted health guidance.
- Access pregnancy health tips and educational content.
- View and manage previous predictions and medical reports.

The application integrates a Machine Learning prediction model, OCR-based medical report analysis, and a secure database into a single user-friendly platform. It is designed to simplify pregnancy monitoring, improve early risk awareness, and provide accessible healthcare support while emphasizing that all AI-generated recommendations should be verified by healthcare professionals.

2. Overall Description

2.1 Product Perspective

The Maternal Care Platform is a standalone AI-powered web application built with Streamlit. It integrates a Machine Learning model, OCR-based medical report analysis, an AI Health Assistant, and a SQLite database to provide pregnancy monitoring and risk assessment through a single, easy-to-use interface.

The system allows users to securely manage their accounts, receive pregnancy risk predictions, monitor pregnancy progress, analyze laboratory reports, communicate with the AI assistant, and review their previous prediction history. All user data is securely stored to provide a personalized experience throughout the pregnancy journey.

The platform is designed to support early pregnancy risk awareness and healthcare monitoring while serving as a supportive decision-making tool rather than a replacement for professional medical advice.

2.2 Product Functions

The Maternal Care Platform provides the following core functionalities:

- User Registration and Authentication

- Personal Dashboard
- Pregnancy Risk Prediction
- Pregnancy Tracker
- AI Health Assistant
- Medical Report Scanner
- Pregnancy History Management
- Health Tips and Educational Content
- Secure User Data Storage

Functional Requirements

Function	Function 1 — User Registration
Input	Full Name, Email Address, Password
Output	User account created successfully
Processing	Validate user information, hash the password using bcrypt, and store the account in the SQLite users table.

Function	Function 2 — User Login
Input	Email, Password
Output	Authenticated session
Processing	Verify the submitted password against the stored bcrypt hash and initialize the session in st.session_state.

Function	Function 3 — Personal Dashboard
Input	Logged-in User
Output	Dashboard summary view
Processing	Retrieve the user's current pregnancy week, most recent prediction, and most recent AI conversation summary.

Function	Function 4 — Pregnancy Risk Prediction
Input	Age, Systolic BP, Diastolic BP, Blood Sugar, and related vitals
Output	Risk Level (Low / Mid / High), Confidence, Recommendation
Processing	Scale the input features and classify them using the trained Random Forest model; store the result in the predictions table.

Function	Function 5 — Pregnancy Tracker
Input	LMP or Estimated Due Date
Output	Current Pregnancy Week, Trimester, Milestones
Processing	Calculate gestational week from the provided date and display relevant week-by-week milestone content.

Function	Function 6 — AI Health Assistant
Input	User's Question (free text)
Output	Matched Answer or Emergency Guidance
Processing	Apply keyword and fuzzy matching against the knowledge base; detect emergency keywords and prioritize safety guidance; persist the exchange to chat_messages.

Function	Function 7 — Medical Report Scanner (Upload & OCR)
Input	PDF, JPG, or PNG lab report
Output	Extracted Raw OCR Text
Processing	Preprocess the image, read an existing PDF text layer directly when available, otherwise run OCR (EasyOCR, falling back to Tesseract).

Function	Function 8 — Medical Value Extraction
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Input	OCR / PDF Text
Output	Structured Medical Values, Missing-Field List
Processing	Apply per-field regular-expression patterns tolerant of different lab abbreviations and formats.

Function	Function 9 — Report Summary & Prediction
Input	Reviewed Medical Values
Output	Plain-language Summary, Risk Prediction
Processing	Generate a plain-language summary of flagged values and run the shared Risk Engine on the confirmed values.

Function	Function 10 — Pregnancy History
Input	User ID
Output	List of Past Predictions and Scanned Reports
Processing	Retrieve, search, filter, and sort historical records from the database; render charts and support CSV export.

2.3 User Classes

The Maternal Care Platform is designed for a single primary user:

- **Registered User (Pregnant Woman)**

The user can create an account, log in securely, predict pregnancy risk, track pregnancy progress, upload medical reports for analysis, communicate with the AI Health Assistant, and view previous predictions and reports. The interface is designed to be simple and user-friendly for users with limited technical experience.

2.4 Operating Environment

Hardware

- Multi-core Processor (Intel Core i5 or equivalent)
- Minimum 8 GB RAM
- Minimum 5 GB Free Disk Space

Software

- Python 3.11+
 - Streamlit
 - Scikit-learn
 - SQLite
 - EasyOCR
 - OpenCV
 - Plotly
 - bcrypt
-

2.5 Design Constraints

The system operates under the following constraints:

- Pregnancy risk prediction is performed using the trained Random Forest model.
 - OCR accuracy depends on the quality of uploaded medical reports.
 - Supported report formats are PDF, JPG, and PNG.
 - User sessions require authentication before accessing system features.
 - AI-generated responses are intended for informational purposes only and do not replace professional medical advice.
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2.6 Project Documentation

The project includes:

- Software Requirements Specification (SRS)
 - README
 - Database Schema
 - ER Diagram
 - Use Case Diagram
 - Activity Diagrams
 - System Architecture
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2.7 User Guide

The platform provides guidance for:

- User Registration and Login
 - Pregnancy Risk Prediction
 - Pregnancy Tracker
 - AI Health Assistant
 - Medical Report Scanner
 - Pregnancy History
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2.8 Assumptions

The system assumes that:

- Users access the platform through a supported web browser.
- Medical reports are uploaded in supported formats.

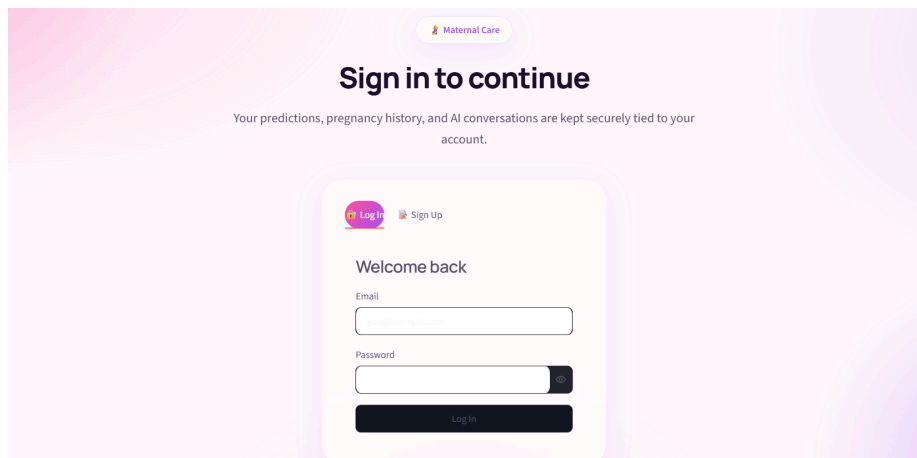
- The trained Machine Learning model and OCR engine are available.
- Users provide accurate health information for reliable predictions.
- AI-generated recommendations are supportive and should always be confirmed by healthcare professionals.

3. External Interface Requirements

3.1 User Interfaces

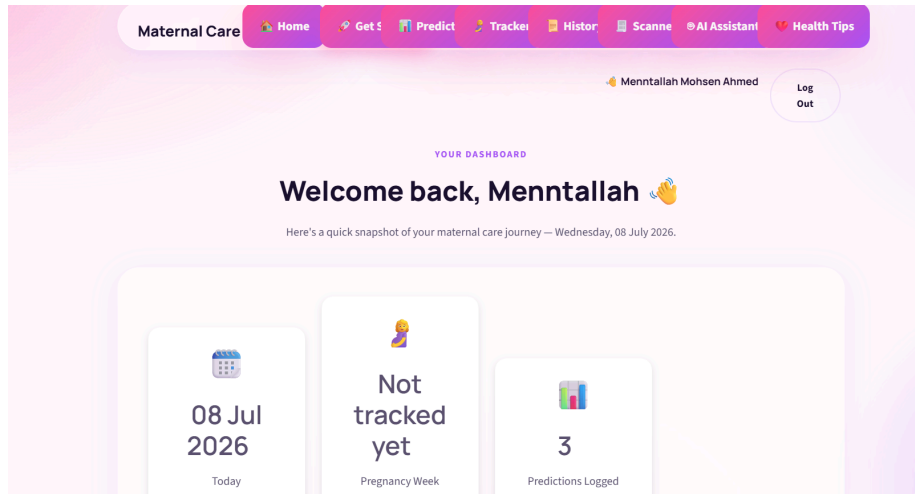
The Maternal Care Platform provides a clean, form-driven Streamlit interface organized as a multipage application, accessible through a sticky top navigation bar (Home, Get Started, Predict, Tracker, History, Scanner, AI Assistant, Health Tips).

Sign In / Sign Up interface is used to create or access a user account with email and password. Every other page is gated behind a successful login.



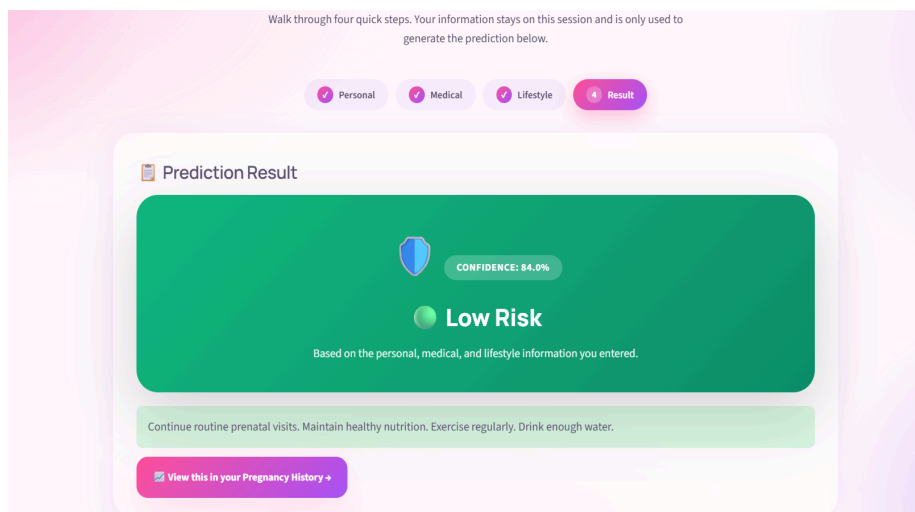
Sign In Page

The Personal Dashboard greets the returning user by name, shows the current date, pregnancy week, and quick stat cards summarizing predictions logged so far.



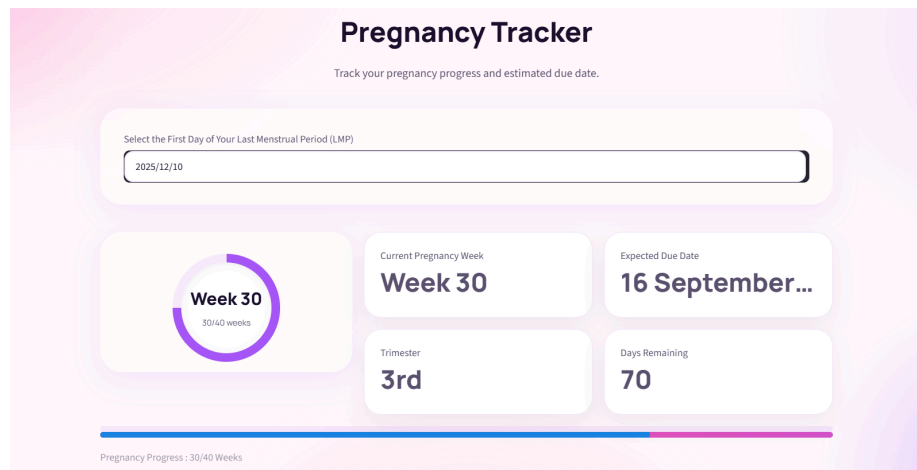
Personal Dashboard

The Risk Prediction page walks the user through a four-step form (Personal, Medical, Lifestyle, Result) and returns a risk level, confidence score, and a plain-language recommendation.



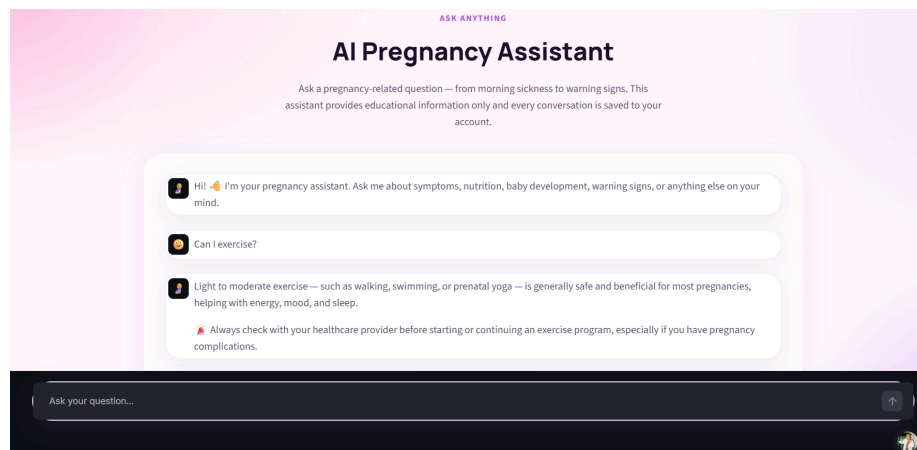
Pregnancy Risk Prediction Result

The Pregnancy Tracker lets the user enter the first day of their last menstrual period and displays the current pregnancy week, trimester, expected due date, and days remaining.



Pregnancy Tracker

The AI Pregnancy Assistant provides a chat-style interface where the user asks a maternal-health question and receives a matched answer from the knowledge base; every conversation is saved to the user's account.



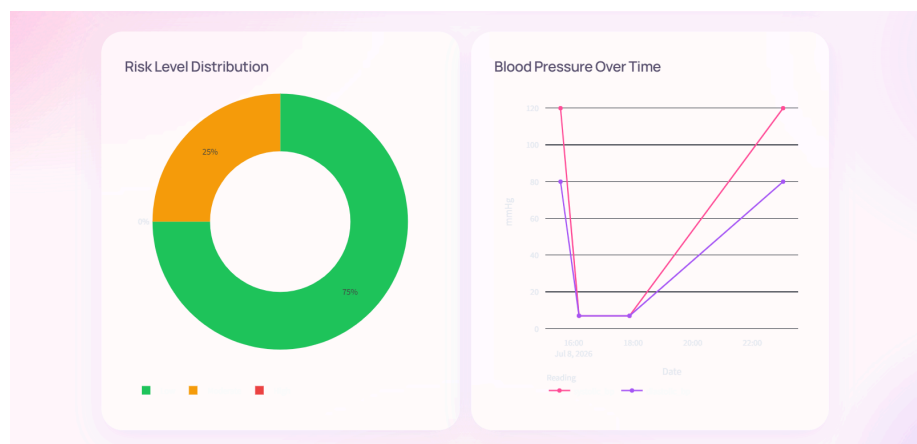
AI Pregnancy Assistant

Health Tips & Risk Awareness offers curated, topic-based educational content (e.g., High Blood Pressure, Gestational Diabetes, Anemia, Obesity, Smoking, Poor Nutrition) with risk explanations and healthy recommendations.



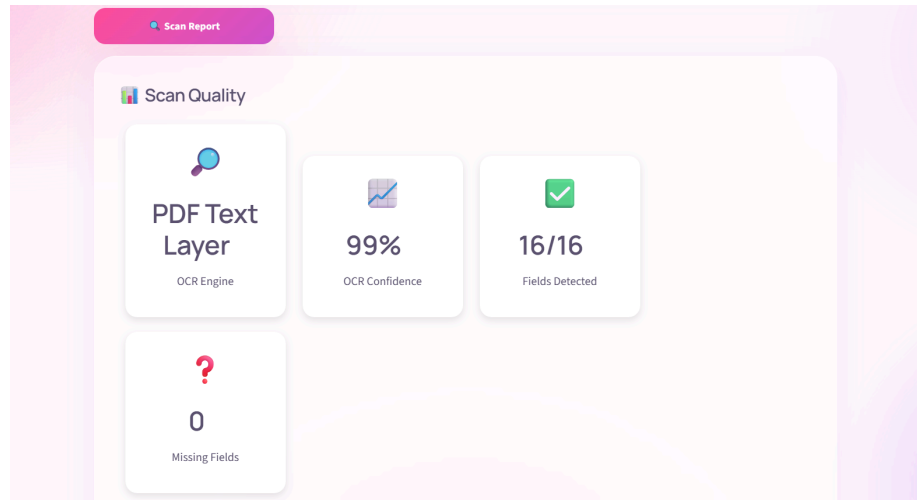
Health Tips & Risk Awareness

The Pregnancy History page renders interactive Plotly charts — a risk-level distribution donut chart and a blood-pressure trend line chart — over the user's saved prediction history.



Pregnancy History Charts

The Medical Report Scanner reports scan quality after processing an uploaded lab report — OCR engine used, OCR confidence, fields detected, and any missing fields — before the extracted values are reviewed and used for prediction.



Medical Report Scanner: Scan Quality

The interface reuses a consistent visual design system defined in `assets/style.css` across all pages, and all forms provide inline validation feedback before submission.

3.2 Communication Interfaces

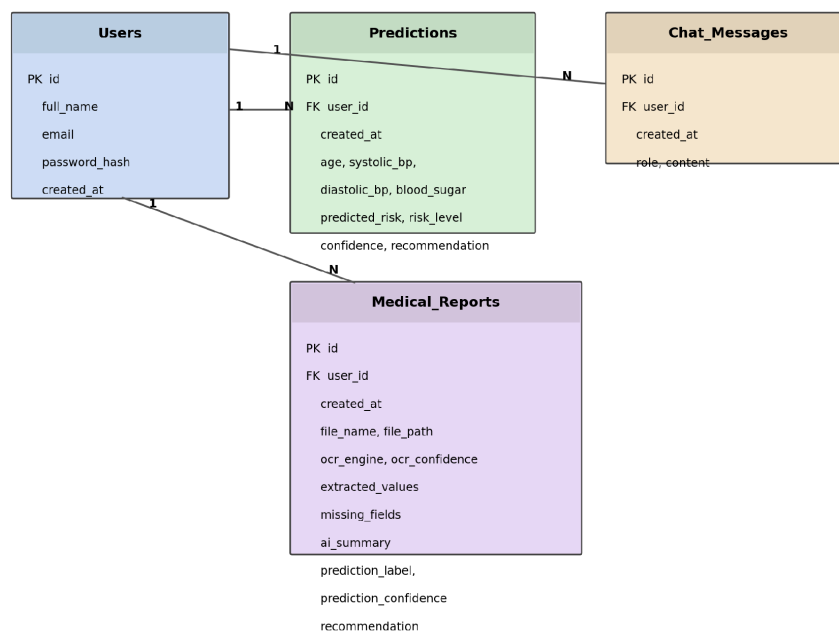
The Maternal Care Platform is a web-based application developed using **Streamlit**. Users access the system through a web browser, where requests are processed by the application and stored in a local **SQLite** database.

All system features require user authentication before access. The Medical Report Scanner performs OCR processing locally using **EasyOCR** and **Tesseract**, ensuring that uploaded medical reports are processed within the application without relying on external services. The platform is designed to provide a secure and responsive user experience while protecting user data.

4. System Features

4.1 E-R Diagram

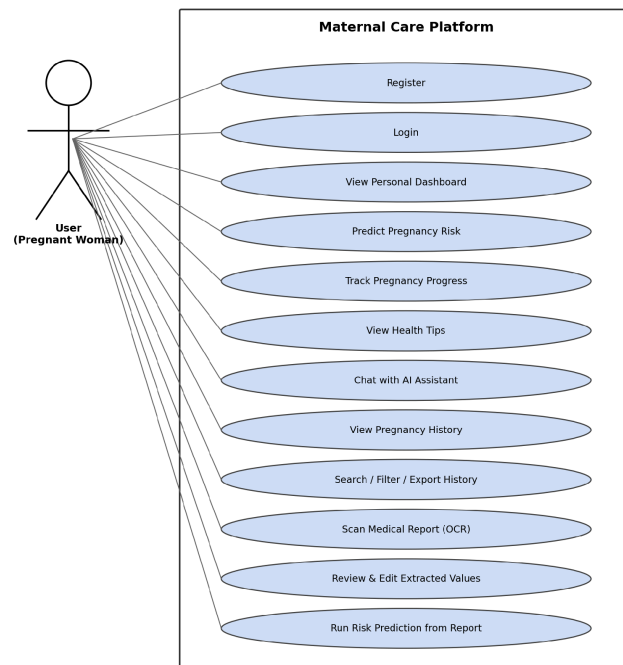
The Entity-Relationship (ER) Diagram illustrates the database structure of the Maternal Care Platform. It consists of four main tables: **Users**, **Predictions**, **Chat_Messages**, and **Medical_Reports**. The **Users** table is linked to the other tables through a one-to-many relationship using **user_id**, allowing each user to store multiple predictions, conversations, and scanned reports.



Entity-Relationship Diagram

4.2 Use Case Diagram

The Use Case Diagram illustrates the interactions between the **Registered User** and the system. The user can register, log in, predict pregnancy risk, track pregnancy progress, communicate with the AI Health Assistant, upload medical reports for analysis, and review previous predictions and reports.



Use Case Diagram

4.3 Use Case Scenarios

1) User Registration

Use Case Name	User Registration
Goal	Create a new account.
Primary Actor	Registered User
Precondition	User is not registered
Postcondition	New account created
Main Flow	1. User opens the Sign Up page.2. Enters full name, email, password, and confirms password.3. System validates the input.4. Password is hashed using bcrypt.5. User account is created in SQLite.6. Success message displayed; user is redirected to Sign In.

2) User Login

Use Case Name	User Login
Goal	Authenticate the user and start a session.
Primary Actor	Registered User
Precondition	User is registered
Postcondition	User is logged in
Main Flow	1. Enter email.2. Enter password.3. Validate credentials against the stored hash.4. Initialize st.session_state for the user.5. Redirect to the Personal Dashboard.

3) View Personal Dashboard

Use Case Name	View Personal Dashboard
Goal	Show a personalized summary on login.
Primary Actor	Registered User
Precondition	User is logged in
Postcondition	Dashboard displayed
Main Flow	1. User lands on the Home page after login.2. System retrieves current pregnancy week (if Tracker has been used).3. System retrieves the most recent prediction.4. System retrieves the most recent AI conversation summary.5. Dashboard is rendered with quick stat cards.

4) Predict Pregnancy Risk

Use Case Name	Predict Pregnancy Risk
Goal	Classify pregnancy risk from manually entered vitals.
Primary Actor	Registered User
Precondition	User is logged in
Postcondition	Risk prediction generated and saved

Main Flow	1. User opens Risk Prediction page.2. User enters vitals (age, blood pressure, blood sugar, etc.).3. System validates input ranges.4. Features are scaled using the saved Scaler.5. Random Forest model predicts the risk level.6. Result and recommendation are stored in the predictions table.7. Result is displayed to the user.
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5) Track Pregnancy Progress

Use Case Name	Track Pregnancy Progress
Goal	Show the current pregnancy week and milestones.
Primary Actor	Registered User
Precondition	User is logged in
Postcondition	Weekly milestones displayed
Main Flow	1. User opens the Pregnancy Tracker page.2. User enters or confirms the LMP / due date.3. System calculates the current week and trimester.4. System displays relevant week-by-week milestone content.5. Dashboard summary is updated.

6) Chat with AI Assistant

Use Case Name	Chat with AI Assistant
Goal	Answer a maternal-health question.
Primary Actor	Registered User
Precondition	User is logged in
Postcondition	Answer displayed and conversation saved
Main Flow	1. User types a question in the AI Assistant chat.2. System applies keyword and fuzzy matching against the knowledge base.3. System checks for emergency keywords.4. If an emergency keyword is detected, emergency guidance is prioritized.5. Best-matching answer is returned.6. Conversation turn is saved to chat_messages.

7) Scan Medical Report

Use Case Name	Scan Medical Report
Goal	Extract lab values from an uploaded report.
Primary Actor	Registered User
Precondition	User is logged in
Postcondition	Extracted values ready for review
Main Flow	1. User uploads a PDF, JPG, or PNG lab report. 2. System preprocesses the image (rotate, denoise, sharpen, crop). 3. System reads the PDF text layer directly, or runs OCR (EasyOCR, then Tesseract fallback). 4. System parses medical values using regex patterns. 5. Extracted values and missing fields are shown to the user for review.

8) Review & Edit Extracted Values

Use Case Name	Review & Edit Extracted Values
Goal	Confirm or correct extracted lab values.
Primary Actor	Registered User
Precondition	Report has been scanned
Postcondition	Confirmed values ready for prediction
Main Flow	1. User reviews each pre-filled or 'not detected' field. 2. User corrects or completes any missing values. 3. User confirms the reviewed values.

9) Run Prediction from Report

Use Case Name	Run Prediction from Report
Goal	Generate a risk prediction from a scanned report.
Primary Actor	Registered User
Precondition	Values have been reviewed
Postcondition	Prediction generated and saved

Main Flow	1. System runs the shared Risk Engine on the confirmed values.2. System generates a plain-language summary of flagged findings.3. Result is saved to medical_reports and predictions.4. Result is displayed to the user.
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10) View Pregnancy History

Use Case Name	View Pregnancy History
Goal	Review past predictions and reports.
Primary Actor	Registered User
Precondition	User is logged in
Postcondition	History displayed
Main Flow	1. User opens the Pregnancy History page.2. System retrieves saved predictions and scanned reports from the database.3. User searches, filters by risk level or date, and sorts records.4. System renders risk-distribution and blood-pressure-trend charts.5. User may export the filtered records to CSV.

4.4 Activity Diagrams

1) User Registration

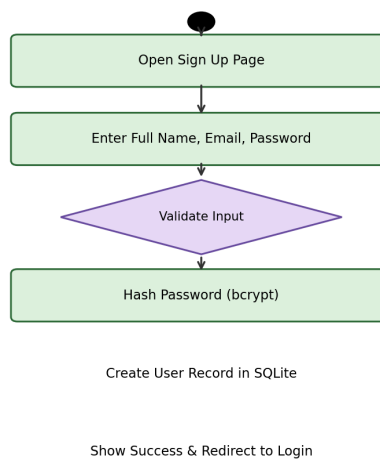
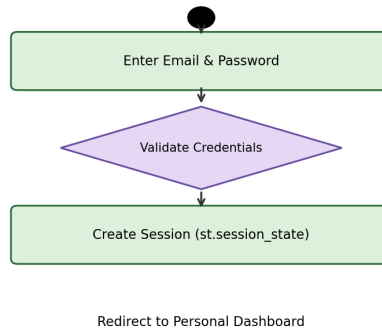


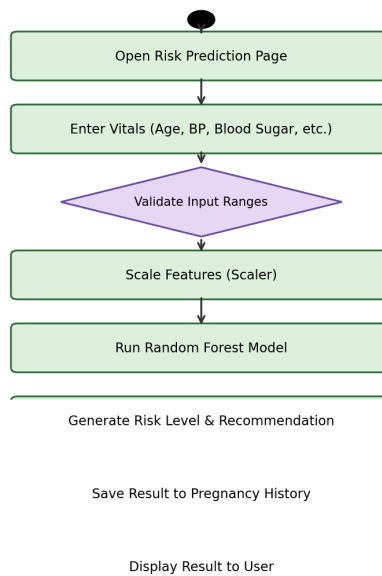
Figure 4.4.1 — User Registration Activity Diagram

2) User Login



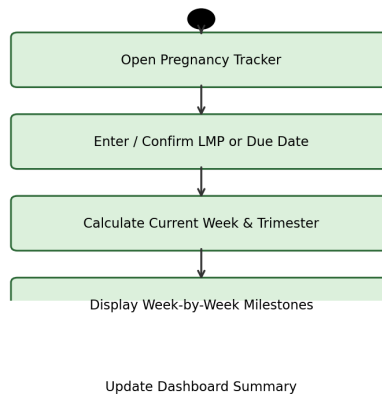
User Login Activity Diagram

3) Pregnancy Risk Prediction



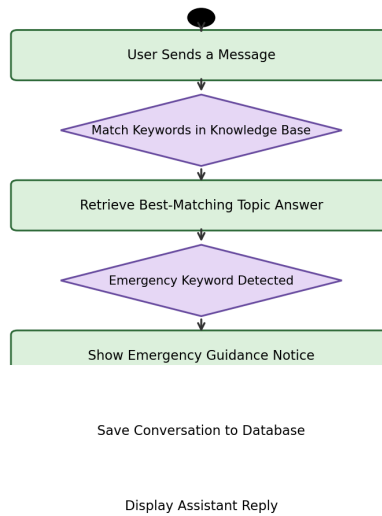
Risk Prediction Activity Diagram

4) Pregnancy Tracker



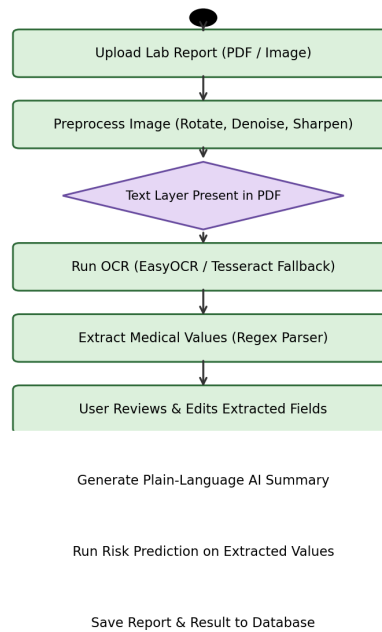
Pregnancy Tracker Activity Diagram

5) AI Health Assistant



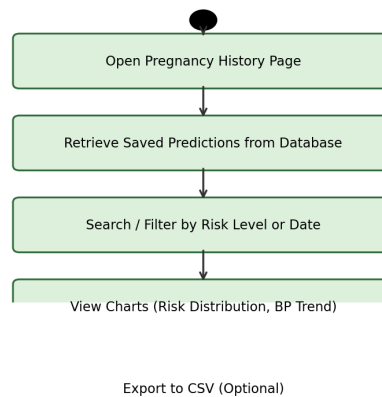
AI Assistant Activity Diagram

6) Medical Report Scanner



Medical Report Scanner Activity Diagram

7) Pregnancy History



Pregnancy History Activity Diagram

4.5 System Architecture

4.5.1 Overall Architecture

The Maternal Care Platform follows a modular architecture built with **Streamlit**. The system integrates user authentication, a Machine Learning prediction model, OCR-based medical report analysis, an AI Health Assistant, and a **SQLite** database within a single web application.

Users interact with the platform through a simple web interface to access prediction, pregnancy tracking, medical report analysis, and AI-assisted guidance. All user data, predictions, conversations, and reports are securely stored in the database.

4.5.2 Risk Prediction Module

The Risk Prediction Module uses a trained **Random Forest** model to analyze maternal health information and classify pregnancy risk into Low, Moderate, or High risk. The same prediction module is used for both manually entered health data and values extracted from medical reports, ensuring consistent results.

4.5.3 Medical Report Scanner Module

The Medical Report Scanner automatically analyzes uploaded medical reports using **OCR** technology.

Its main responsibilities include:

- Processing uploaded PDF and image files.
 - Extracting medical information using OCR.
 - Identifying the required health measurements.
 - Allowing users to review extracted values before prediction.
 - Sending the extracted data to the Risk Prediction Module.
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4.5.4 AI Health Assistant Module

The AI Health Assistant provides pregnancy-related guidance through a predefined knowledge base.

Its responsibilities include:

- Answering pregnancy-related questions.
 - Providing health recommendations.
 - Detecting emergency-related keywords and displaying appropriate guidance.
 - Saving conversation history for each user.
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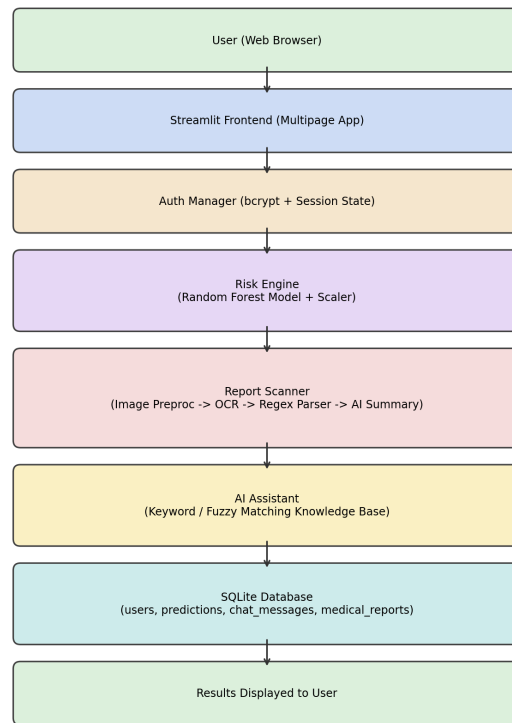
4.5.5 Database Design

The platform uses **SQLite** to store all application data.

The database contains the following main tables:

- Users
- Predictions
- Chat_Messages
- Medical_Reports

These tables store user information, prediction history, AI conversations, and scanned medical reports while maintaining relationships between all user records.



Architecture Flow

5. Other Non-Functional Requirements

5.1 Performance Requirements

The Maternal Care Platform shall provide a fast and responsive user experience. The system should:

- Authenticate users efficiently.
 - Generate pregnancy risk predictions within a short response time.
 - Process medical reports and OCR analysis with acceptable performance.
 - Display dashboard data and pregnancy history without noticeable delay.
 - Support multiple user sessions while maintaining stable performance.
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5.2 Safety Requirements

To ensure safe and reliable operation, the platform shall:

- Require user authentication before accessing protected features.
 - Accept only supported medical report formats.
 - Validate extracted medical values before prediction.
 - Display reminders that AI-generated recommendations do not replace professional medical advice.
 - Detect emergency-related queries and prioritize appropriate guidance.
 - Handle unexpected errors gracefully without affecting user experience.
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5.3 Security Requirements

The platform shall provide:

- Secure password hashing using **bcrypt**.
 - Session-based authentication.
 - Input validation for all user forms.
 - Secure storage of user information and prediction history.
 - Local processing of uploaded medical reports to protect user privacy.
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5.4 Software Quality Attributes

The Maternal Care Platform is designed to provide:

- **Availability:** Accessible whenever the application is running.
 - **Reliability:** Stable operation under normal usage.
 - **Usability:** A simple and intuitive interface.
 - **Maintainability:** Modular architecture for easier updates.
 - **Scalability:** Support for future enhancements.
 - **Portability:** Compatibility with Windows, Linux, and macOS.
 - **Data Integrity:** Consistent and secure storage of user information.
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5.5 Business Rules

The system follows these business rules:

1. Only registered users can access protected features.
2. Every pregnancy risk prediction is automatically saved to the user's history.
3. Medical reports must be uploaded in supported formats (PDF, JPG, or PNG).
4. Extracted medical values should be reviewed before prediction.
5. AI-generated recommendations are for informational purposes only and do not replace professional medical advice.
6. Users can access only their own data and prediction history.

6. Other Requirements

The project includes the following documentation:

- Software Requirements Specification (SRS)
- README
- Database Schema
- ER Diagram
- Use Case Diagram
- Activity Diagrams
- System Architecture

The platform requires:

- A supported web browser.
- A valid user account.
- Python 3.11+ with the required project dependencies.

Appendix A: Glossary

Term	Description
ANC	Antenatal Care — routine checkups during pregnancy.
BP	Blood Pressure, expressed as systolic/diastolic values.
OCR	Optical Character Recognition — technology used to read text from images or scanned documents.
Random Forest	An ensemble Machine Learning algorithm used for the pregnancy-risk classification model.
Scaler	A StandardScaler object used to normalize input features before they are passed to the model.
EasyOCR	The primary OCR engine used by the Medical Report Scanner.
Tesseract	The fallback OCR engine used when EasyOCR is unavailable.
SQLite	A lightweight, file-based relational database used for all application data.
bcrypt	A password-hashing algorithm used to securely store user passwords.
Trimester	One of the three approximately three-month periods of a pregnancy.
Knowledge Base	The curated set of maternal-health topics and answers used by the AI Assistant.
LMP	Last Menstrual Period — used as a reference point for calculating gestational age.